

KEYFACTOR

Shaping the Future of IAM: How to Prepare for Four Disruptive Changes

How Machine Identities (and the Cryptography Behind Them) are Changing Fast



Chris Hickman

Chief Security Officer at Keyfactor

The Perfect Storm is Approaching



KEYFACTOR

1

AI and workload identity

Rapidly increasing machine identity and cryptographic surface area that must be implemented and managed

2

Shorter lifespans

Shifting to 47-day TLS certificates by 2029, putting increased operational pressure on teams

3

Regulatory changes

A new wave of regulations and standards requiring cryptographic hygiene—DORA, CRA, NIS 2, PCI DSS 4.0

4

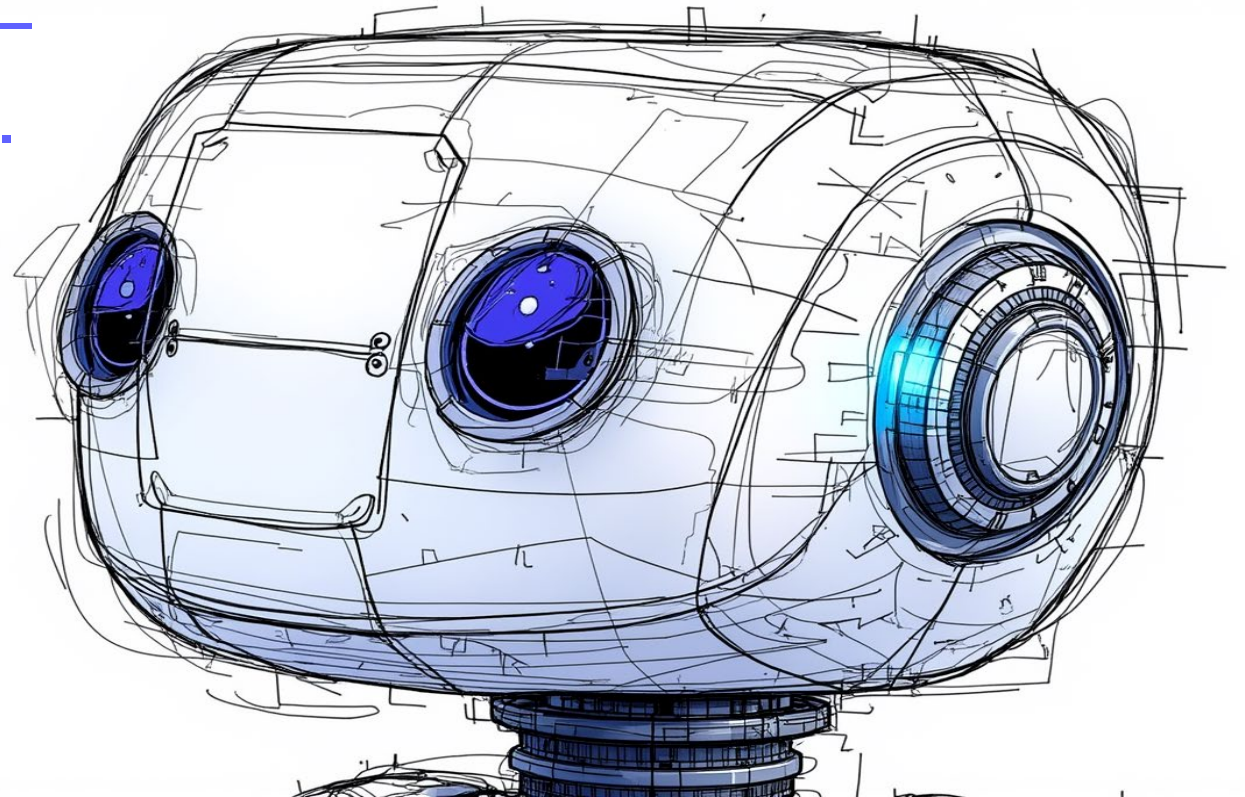
Quantum risks

A complete infrastructure transformation and a fast-closing window for migration (2030-2035)

Workloads

Every workload needs an identity —
and every identity must be trusted.

KEYFACTOR



Disruption #1:

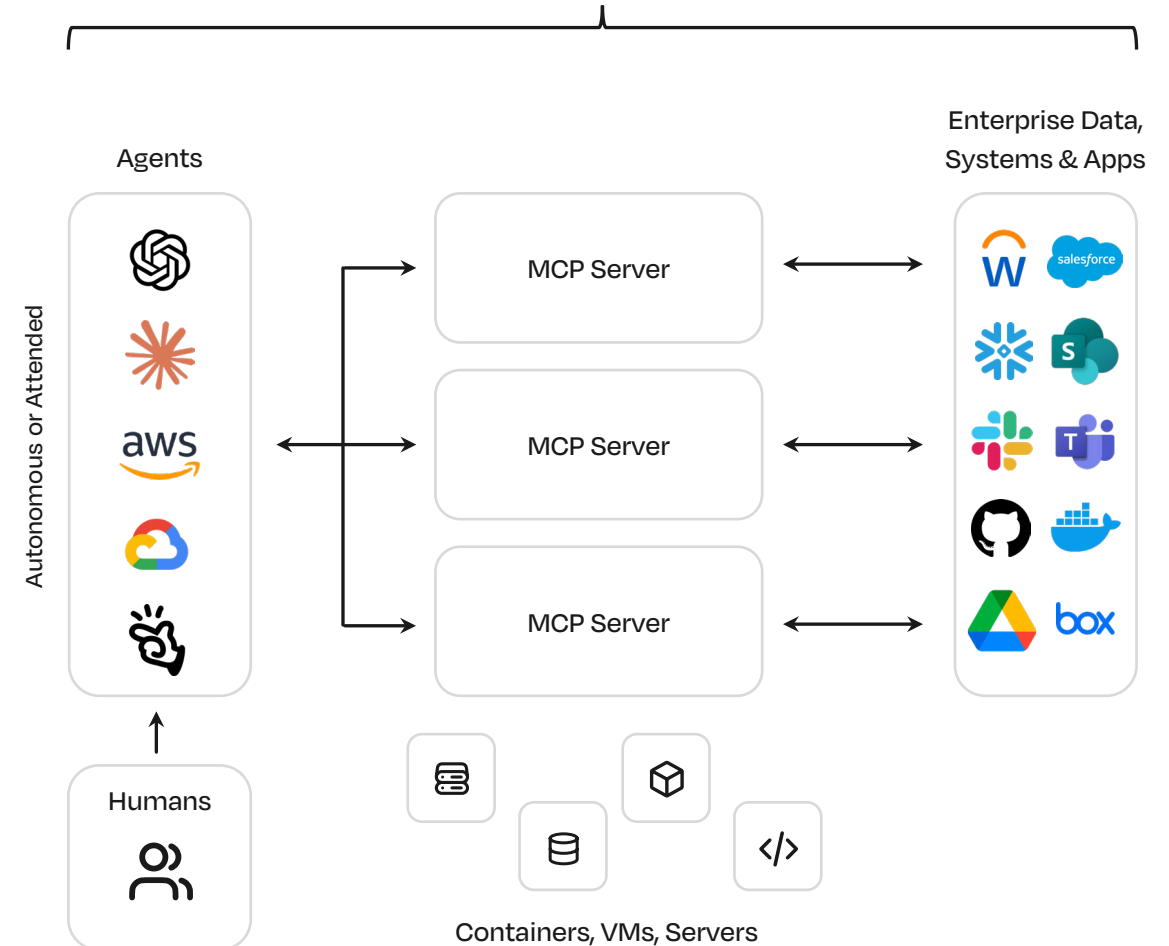
Workloads & AI Agents

92%

Of companies plan to increase AI investments over the next three years

Superagency in the workplace:
Empowering people to unlock AI's full potential, McKinsey

Every human, agent, and workload needs a **unique identity**—or it cannot be trusted



KEYFACTOR

Zero Trust for AI & Workloads

A framework for success



Assign every AI agent an X.509 certificate as its unique digital identity



Enable mutual TLS for agent-to-system and agent-to-agent communications

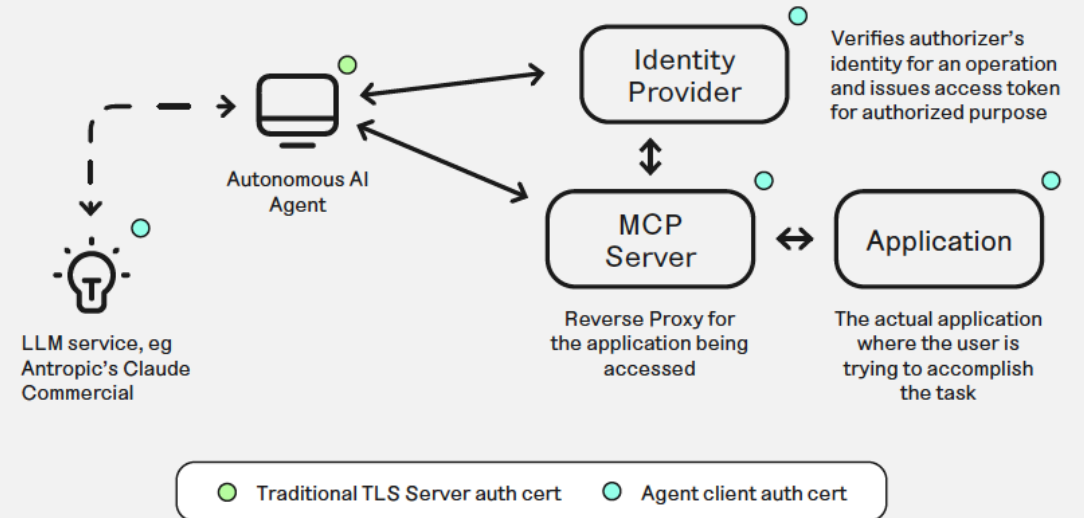


Manage certificates automatically through standard or non-standards-based automation



Extend this model to workloads across your cloud environments (containers, VMs, etc.)

MCP Security - Autonomous OAuth





Shorter Lifespans

Digital trust now expires every 47 days.

Disruption #2:

Shorter TLS Lifespans



Kwantem • 7d ago

F@C* it I'm retiring

↑ 12 ↓ Reply



b1oHeX • 6d ago

Oh my lanta, I just inherited PKI too

↑ 1 ↓ Reply Award Share ...



juicefarm • 7d ago

Might as well make them expire after 1 second at this point if this is the guiding logic. You want to get nuts. LETS GET NUTS!!

↑ 121 ↓ Reply Award Share ...

+ 15 more replies

398 days



Today

200 days



2026
March 15

100 days



2027
March 15

47 days



2029
March 15

10-12x

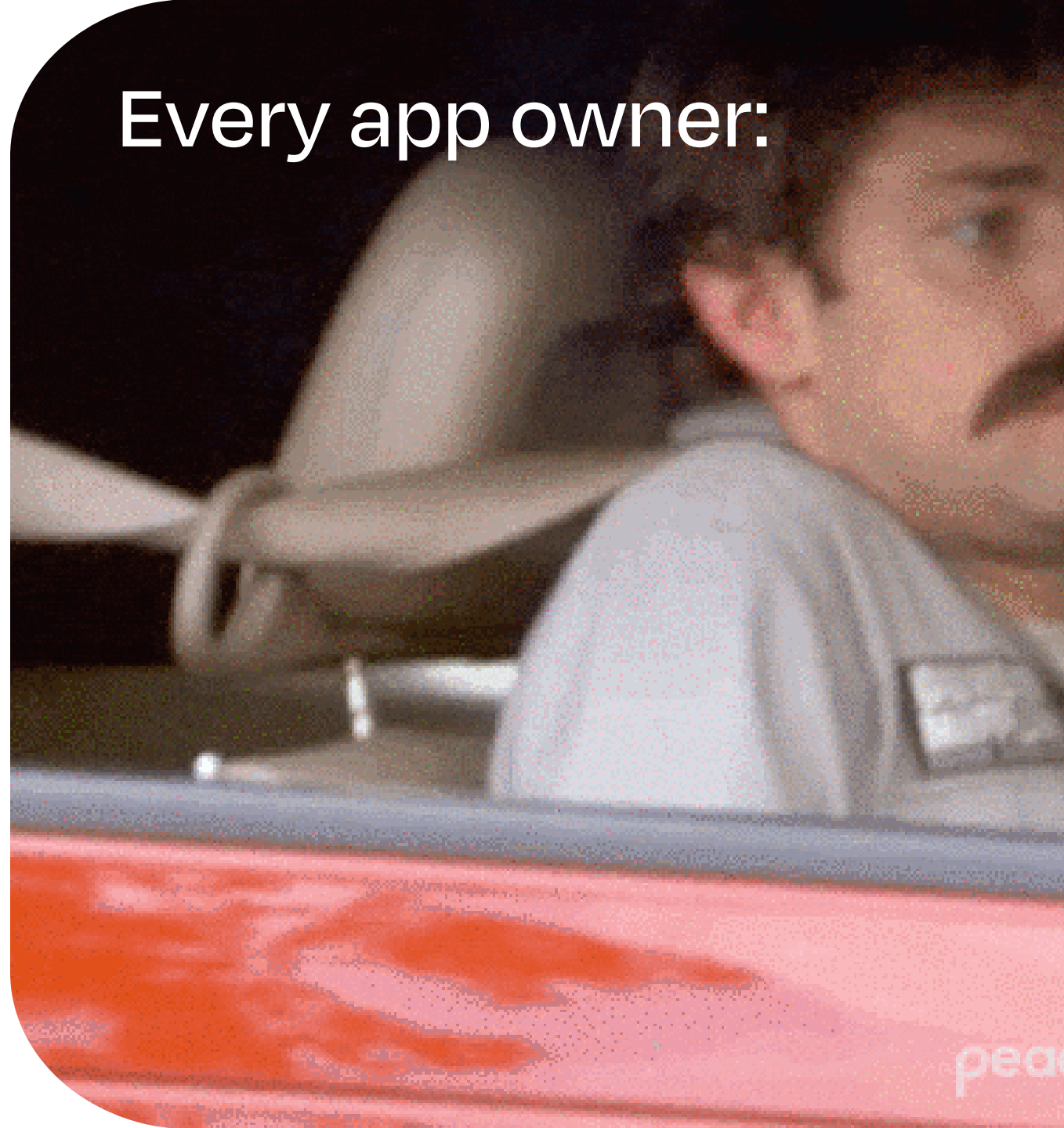
Increase in workload with
47-day certificate lifespans

KEYFACTOR

Cert team:

"Who forgot to
renew their cert?!"

Every app owner:

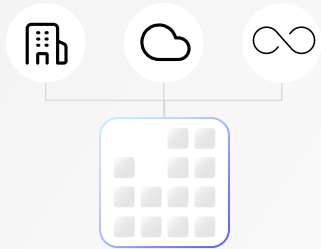


Key ingredients to getting it right

Hot tip:

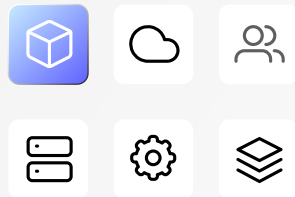
Identify all the apps and teams that use publicly-trusted certificates and where they are not necessary – this can reduce cost and operational overhead.

Automation at every step



Visibility

Build a centralized, complete, and real-time inventory of certificates



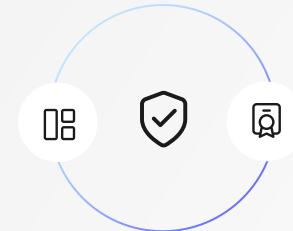
Accountability

Assign certificate ownership to BUs, teams, or individuals



Monitoring

Continuously monitor inventory and set alerts for renewals and other issues



Workflows

Simplify request and approval workflows to reduce error and friction

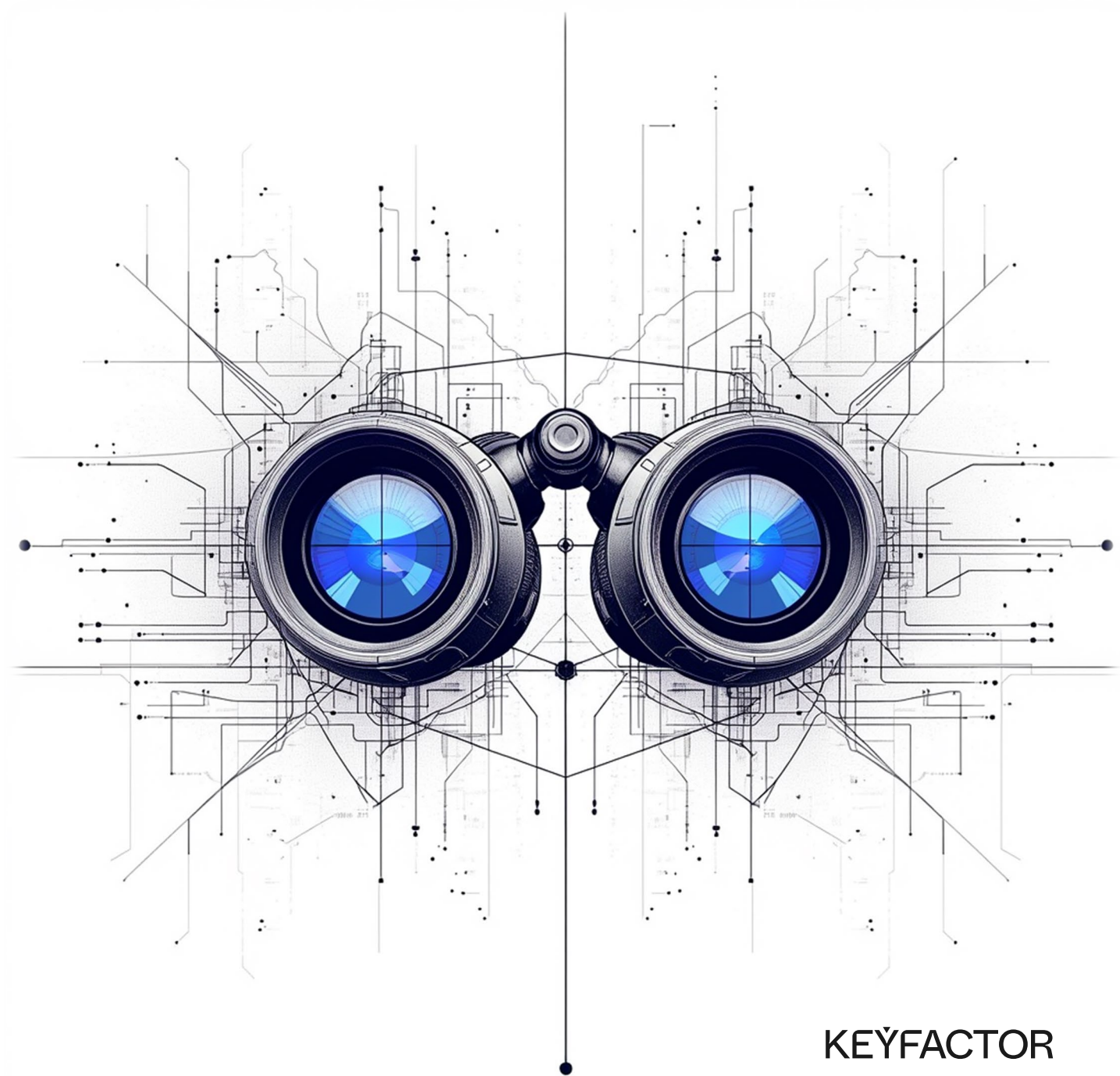


Deployment

Determine best methods to automate renewal and provisioning (flexibility)

You can't migrate
what you can't see

Quantum

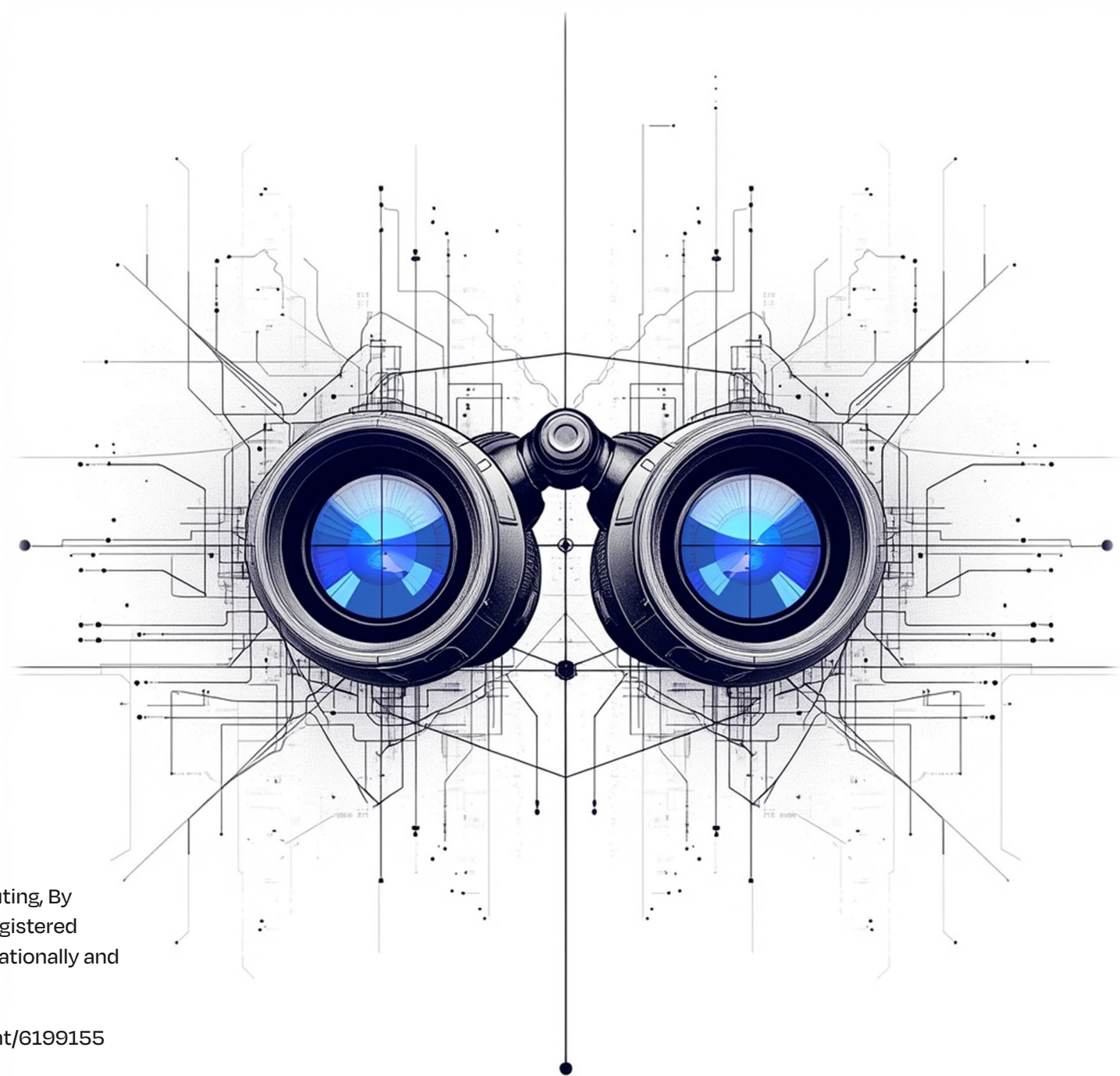


KEYFACTOR

According to Gartner[®], "By 2029, advances in quantum computing will make conventional asymmetric cryptography unsafe to use".

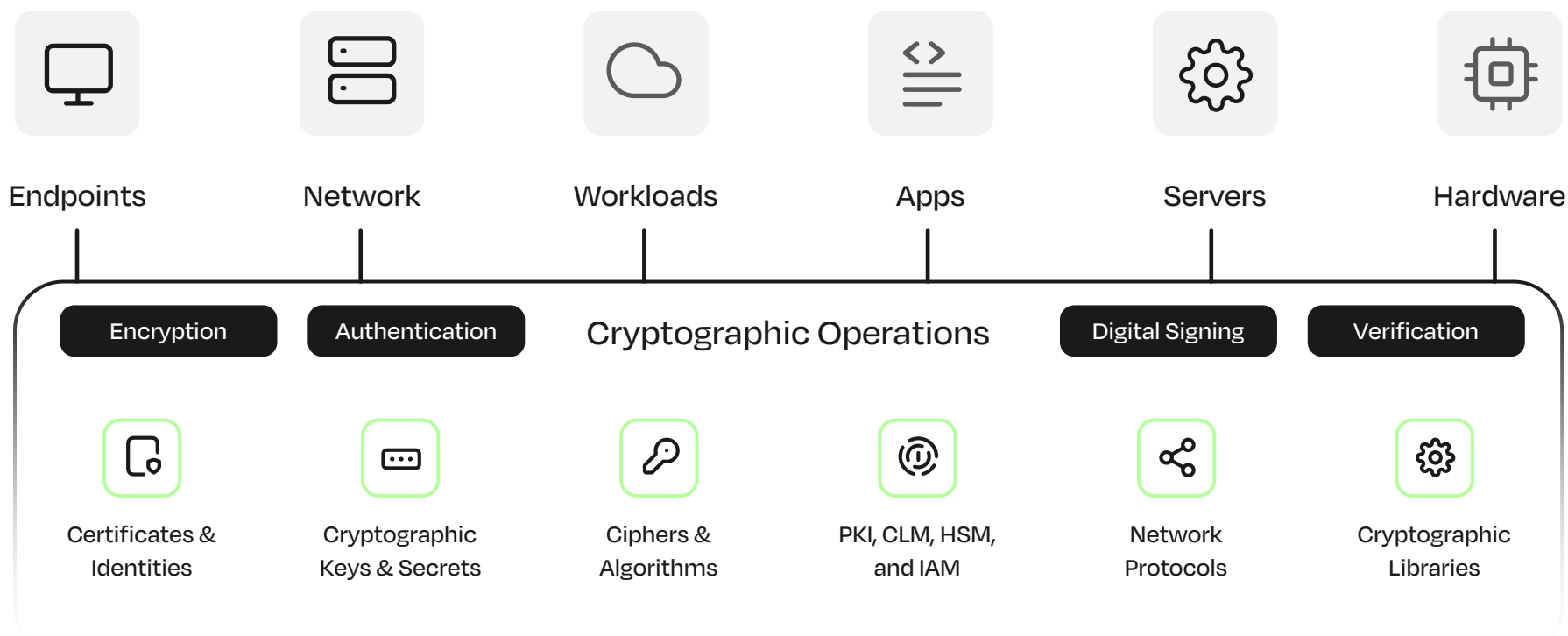
Source: Gartner, How to Mitigate the Cryptography Risks Posed by Quantum Computing, By Thomas Lintemuth, Craig Lawson, Mark Horvath, 21 February 2025. GARTNER is a registered trademark and service mark of Gartner, Inc. and/or its affiliates in the U.S. and internationally and is used herein with permission. All rights reserved.

Gartner subscribers can access the report here: <https://www.gartner.com/document/6199155>



The Reality

Cryptography is Critical Infrastructure



KEYFACTOR

Steps to Quantum Readiness

Many companies are around here

01 Inform & Educate

- Engage senior stakeholders
- Bring a risk-based narrative (operational, security, compliance)
- Budget requires buy-in

02 Discover

- Discovery and mapping of cryptographic assets
- Source code, file systems, endpoints, servers and hosts, network interfaces
- All cryptographic assets

03 Inventory & Assess

- Inventory is CONTINUOUS
- Assess the vulnerability and dependencies of cryptographic assets
- Prioritize by risk: High, Med, Low

04 Test

- Begin testing quantum-resilient algorithms in practice
- Plan for migration of weak/outdated cryptography

05 Upgrade Infrastructure

- Upgrade your underlying infrastructure (HSM, KMS, CLM, PKI, signing, cryptographic libraries)

07 Continuously Manage

- Continuously monitor and manage your enterprise cryptography through policy and processes

06 Begin Rollout

- Begin to implement NIST
- PQC algorithms into production based on your prioritization

Talk to your vendors!

Assess the quantum-readiness of technology vendors and supply chain partners

Disruption #4:

Compliance Pressure

US:

- HIPAA
- NIST IR 8547
- CNSA 2.0
- OMB-23-02

EU:

- GDPR
- NIS 2
- DORA
- CRA
- EU Commission PQC Roadmap

UK:

- NCSG

Global:

- PCI DSS 4.0

- ASD

96%

Of organizations are spending more on cybersecurity due to regulations over the last 12 months



PwC 2025 Global Digital Trust Insights Survey

KEYFACTOR

Example:

PCI DSS 4.0

Scope:

Affects any organization or service provider that stores, processes, or transmits payment card data (retailers, banks, FinServ, etc.).

Recent change:

- Introduced March 31, 2022
- Mandatory compliance by March 31, 2025

New Requirement 4.2.1

Organizations must confirm that certificates used over open, public networks are valid, and not expired or revoked

New Requirement 4.2.1.1

Organizations must keep an up-to-date inventory of keys and certificates

New Requirement 12.3.3

Organizations must document and review cryptographic cipher suites and protocols in use at least once every 12 months

Why a Cryptographic Inventory Matters

Organizations face
tightening regional and
industry-specific
regulations requiring
cryptographic visibility

PCI DSS v4.0

Requires inventory of cipher suites, keys, and certificates.

NIST / FIPS 140-3

Mandates validated cryptography tracking.

EU PQC Roadmap & OMB M-23-02

Inventories of quantum-vulnerable systems.

CMMC / FedRAMP

Requires visibility into cryptographic modules.

GDPR & ISO 27001

Strong encryption controls and auditability.

KEYFACTOR

The Perfect Storm is Approaching



KEYFACTOR

1

AI and workload identity

Rapidly increasing machine identity and cryptographic surface area that must be implemented and managed

2

Shorter lifespans

Shifting to 47-day TLS certificates by 2029, putting increased operational pressure on teams

3

Regulatory changes

A new wave of regulations and standards requiring cryptographic hygiene—DORA, CRA, NIS 2, PCI DSS 4.0

4

Quantum risks

A complete infrastructure transformation and a fast-closing window for migration (2030-2035)

Thank you,

for your attention and
for your time.



Chris Hickman

Chief Security Officer at Keyfactor

Curious & want
to learn more?

KEYFACTOR

Booth #606



Your Go-To Hub for
Keyfactor Resources

KEYFACTOR